Robotic Assembly of Deformable Linear Objects via Curriculum Reinforcement Learning

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***Abstract*—The automated assembly of flexible objects presents significant challenges. Although significant progress has been made in the assembly of rigid objects, the methods used for rigid objects cannot be directly applied to flexible objects due to their infinite degrees of freedom. This study proposes a reinforcement learning (RL) based method for deformable cable insertion tasks executed with a universal 2-finger gripper. Firstly, a vision-based detection method is employed to monitor the cable's state in real time, while a state classifier is introduced to provide real-time reward feedback for RL training. Secondly, an adaptive curriculum learning (CL) method is proposed to adjust the initial degree of cable bending through the success rate in the training process, allowing the RL agent to learn progressively from easier to more difficult tasks. The validation experiments were conducted on a type-C cable insertion task, where the robot grips the cable portion of the electrical connector. The results indicate that our method is capable of adapting to various degrees of cable bending, successfully handling cable configurations bent up to a maximum of 40° from its straight, unbent state, with an assembly success rate of over 90%.**

***Index Terms*—Deformable linear objects, robotic assembly, curriculum learning.**

I. Introduction

ASSEMBLY stands as a pivotal stage within the manufacturing process. The peg-in-hole (PiH) task, a type of assembly task, has been widely researched in recent years. However, the majority of research focuses on the assembly of rigid objects while overlooking deformable ones.

Deformable linear objects (DLOs), such as cables, wires, etc., find extensive applications in various fields, such as consumer electronics, automotive, and aerospace industries. Unlike rigid objects, the infinite degrees of deformation freedom inherent in DLOs makes assembly tasks significantly more challenging. Most studies [1], [2] focus on achieving the assembly of connectors and sockets by grasping rigid connectors to avoid cable deformation. While this approach is feasible, practical assembly scenarios may present challenges. For example, numerous low-voltage and high-voltage cables require mating and assembly in the production process of power batteries for electric vehicles. These cables differ in connector interfaces, dimensions, flexibility, and rigidity. Additionally, the installation space for cables is highly constrained. As a result, the assembly of cables has been handled exclusively by human workers. During attempts to automate the plug-in process using robots, it was observed that robotic grippers cannot directly grasp and assemble the connectors in certain positions where assembly spaces are excessively narrow. This highlights the critical importance of developing flexible cable grasping and assembly technologies. Additionally, equipping robot end-effectors with multiple cable clamping mechanisms to accommodate different cable types would increase the end-effector's bulk, thereby limiting operations in tight spaces. Consequently, implementing a universal grasping mechanism capable of handling diverse cables in confined spaces presents both significant application value and notable technical challenges.

RL has demonstrated excellent performance in assembly tasks and has been extensively explored in academic research [3]. However, it still faces significant challenges when it comes to DLOs. First, modeling DLOs in simulation is challenging, resulting in discrepancies between simulated and real-world performance [4]. Additionally, training RL models on real robots is highly inefficient due to the limited efficiency of real training data. Second, the dynamic variation in connector pose makes the design of dense reward functions challenging, which is crucial for the effectiveness of RL.

Humans can manipulate DLOs with their hands and by utilizing contact with the environment to regulate the object's shape. Inspired by this insight, we propose an RL-based framework that utilizes the contact between cables and the environment to adjust the cables' pose and achieve assembly. We introduce expert demonstrations and CL to address the challenges of designing reward functions and the low data efficiency in RL. This approach enables the RL agent to quickly develop strategies for simple tasks under relatively sparse reward functions and gradually tackle more difficult ones. Ultimately, the agent achieves cable assembly with a cable bend angle of up to 40°. The main contributions of this study are listed as follows:

Received 5 January 2025; accepted 8 March 2025. Date of publication 21 March 2025; date of current version 3 April 2025. This article was recommended for publication by Associate Editor C. C. Smith and Editor A. Banerjee upon evaluation of the reviewers' comments. This work was supported in part by the Fundamental Research Funds for the Central Universities under Grant 2024ZYGXZR107 and in part by the GJYC program of Guangzhou under Grant 2024D03J0005. *(Corresponding author: Kai Wu.)*

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This article has supplementary downloadable material available at https://doi.org/10.1109/LRA.2025.3553676, provided by the authors.

Digital Object Identifier 10.1109/LRA.2025.3553676

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1) A RL-based method for assembling flexible cable connectors is proposed, enabling the bent cable to adapt to environmental contact forces compliantly and be smoothly inserted into the socket.
2) A vision-based approach is utilized for real-time monitoring of the cable's state, and an assembly state classifier is proposed to obtain real-time training reward feedback for RL training.
3) An adaptive CL method is proposed, automatically adjusting the initial cable bend angle based on success rates, thereby improving training efficiency and learning success rates.
4) A type-C cable insertion experiment was conducted in real-world experiments, verifying the effectiveness of the proposed method.

II. Related Work

A. Robotic Assembly Tasks With Reinforcement Learning

RL is frequently trained in simulation due to its high data efficiency and the absence of safety risks during contact-rich tasks [5]. Kozlovsky et al. [6] Beltran-Hernandez et al. [2] used RL to determine impedance control parameters in simulation, which were then successfully applied to real robots. Wang et al. [7] verified the feasibility of the strategy in a simulated environment before retraining it in a real-world setting. In contrast, Zhang et al. [8] used domain randomization techniques to eliminate the need for retraining. Ahn et al. [9] achieved rapid adaptation to new scenarios by fine-tuning and replacing the encoder layer responsible for feature extraction.

Sim2real approaches are effective in assembly tasks but struggle with DLOs due to the challenge of modeling their deformation accurately [4]. The strategies developed in the simulation environment may be inapplicable to real-world scenarios. While some studies use simulation data for DLO shape control [10], [11], they typically assume restrictive properties, such as purely elastic deformation.

For contact-rich tasks, real-world training often yields better results, leading some studies to opt for training directly on physical robots. Zhao et al. [12] proposed a learning-based two-stage method **(LBTSM)** and designed a low-dimensional visual state representation. Compared to using the entire image as input, it requires fewer network parameters and leads to more efficient training. Yan et al. [13] utilized human demonstration data to derive an initial policy through Gaussian mixture regression, thereby improving the initial success rate of the network policy. Schoettler et al. [1] employed residual reinforcement learning to accelerate the training process.

Most studies focus on the assembly of rigid objects and assume a fixed grasping pose for the robot-held plug, ignoring the variation in the relative position between the robot and the plug. As a result, these approaches are challenging to apply to DLOs directly. In contrast, our study considers this relative relationship, enabling the robot to handle assembly tasks involving DLOs.

B. Deformable Linear Object Assembly

Handling deformable objects such as fabrics, wires, and cables has always been challenging for robots [4], [14]. Current research typically focuses either on precise observation or modeling of DLOs to accomplish the assembly task. Chang et al. [15] proposed a shared control method for collaborative human-robot plug tasks. They defined the connector pose by extracting 2D cable contours from images and mapping them to 3D space, enabling assembly. De Gregorio et al. [16] perform assembly by extracting the cable terminal contour through thresholding and aligning the connector based on its orientation. Using a shape-based 3D matching method, Zhou et al. [17] match the connector's CAD model to obtain its 6D pose. However, due to the precision limitations of visual perception, it is often challenging to complete tasks in tight-fitting assemblies using vision alone [3]. Wilson et al. [18] use tactile sensing to guide the connector during insertion, ensuring precise alignment. Jasim et al. [19] model the contact states in the assembly of flexible rubber parts using a Gaussian mixture model. Luo et al. [20] integrate force-torque information into RL to tackle the insertion of rigid parts into nonlinearly deformable objects. Blanco-Mulero et al. [21] propose a framework for deformable object manipulation, helping robots understand deformation and interaction to improve control strategies.

Existing methods rely on static initial poses, ignoring pose changes during assembly or depending on precise contact states for successful execution. Our method dynamically adjusts strategies based on the real-time shape of the cable, ensuring greater adaptability and robustness in dynamic tasks.

C. Robotic Assembly Task With Curriculum Learning

CL, which simulates the human learning process by progressively training the model from simple to complex tasks, offers advantages such as accelerated convergence, improved performance, enhanced robustness, and reduced risk of local optima. It has achieved commendable results in robotic assembly tasks. Zhao et al. [12] proposed **LBTSM** that increased the curriculum's difficulty by gradually enlarging the pose error between the robot end-effector and the hole. Wang et al. [7] dynamically adjusted the hole's positioning error and the robot's exploration range based on the success rate. Dong et al. [22] designed CL framework for robotic assembly, where environmental complexity in assembly scenarios and target object diversity were strategically escalated through progressive difficulty levels.

Unlike existing methods, our study employs the degree of cable bending as the curriculum. By designing a reset mechanism, the degree of cable bending is automatically adjusted based on the task success rate, enabling adaptive difficulty adjustment. Our approach allows models to handle assembly tasks with bent cables.

III. Task Description

This study addresses the task of inserting an electrical connector into a socket using a universal two-finger gripper. Unlike

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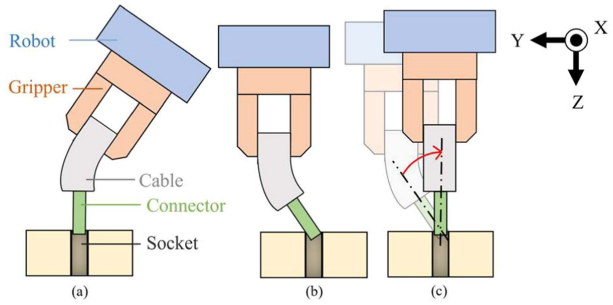


Fig. 1. Strategy for assembly tasks. (a) Connector and socket alignment often require force for insertion due to the transition fit. Additionally, cable bending can obstruct the process. (b) Cable bending causes jamming and potential damage. (c) Using interaction to straighten the cable facilitates alignment and insertion.

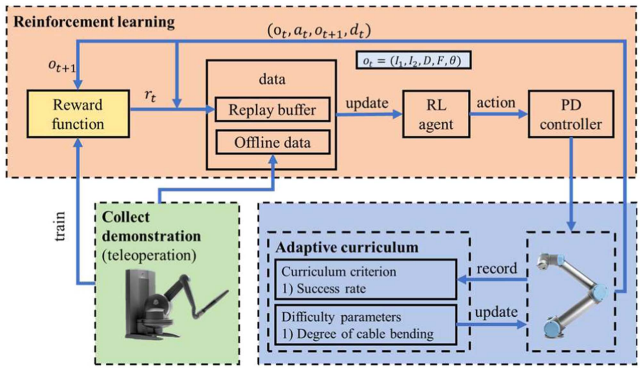


Fig. 2. System pipeline for the insertion task. The green section involves collecting teleoperated demonstrations for offline RL training and classifier development. The blue section adjusts task difficulty online based on success rate. The orange section integrates online and offline RL training, with a PD controller executing policy actions to control the robot.

traditional task setups, the robot's gripper does not grasp the rigid connector directly but instead grips the cable attached to the connector. This allows for a certain degree of cable bending flexibility. As shown in Fig. 1, the cable is initially bent along the x-axis in our task setup, which makes controlling its bending crucial for completing the task. As shown in Fig. 1(a), although the connector and socket are aligned, the fit between them is typically a transition fit, requiring sufficient force to complete the insertion. Even if the robot moves toward the socket, the cable tends to bend further, making it difficult to apply force effectively to the connector. As shown in Fig. 1(b), although the robot's z-axis aligns with the socket, the bending of the cable can easily cause jamming. In such cases, moving the robot toward the socket fails to complete the task and may damage the cable. Therefore, this study involves aligning the connector with the socket and controlling the shape of the cable (Fig. 1(c)), thereby enabling successful insertion.

This study is based on the following assumption: the direction of the socket is known, the robot and the gripper are positioned perpendicular to the socket, the cable is held at the center of the gripper, and the cable is bent along the x-axis. However, the extent of the bending and the position of the socket are unknown.

IV. Method

As discussed in Section III, the cable must be straightened through the interaction between the connector and socket before insertion. The assembly strategy involves two steps: first, locate the socket and partially insert the connector; second, use the connector-socket contact to straighten the cable and complete the insertion. The framework for the robotic assembly of cable is schematically represented in Fig. 2. The proposed method mainly consists of the following three parts: i) A CL method that adaptively adjusts difficulty level based on task success rate; ii) A RL observation space specifically designed for cable assembly tasks; iii) A classifier-augmented reward function. The specific implementation is described as follows.

A. Adaptive Curriculum Learning

In our experimental setup, the curriculum is defined by the initial degree of cable bending, which increases as the task difficulty level rises. The design of this curriculum is motivated by two factors. First, insertion is relatively simple when the cable is nearly straight, requiring only alignment and a direct push. However, as the initial degree of bending increases, the robot must first straighten the cable to ensure proper alignment, adding complexity to the task. Second, a greater initial degree of cable bending causes the connector to tilt more, which increases the cross-sectional area of the connector that enters the socket. This, in turn, reduces the clearance between the connector and the socket, further complicating the insertion process. Consequently, the increase in the initial cable bending directly correlates with an elevation in task difficulty, which is a key aspect of the curriculum design.

The curriculum difficulty is represented as a coefficient ranging from 0 to 1, which adjusts dynamically based on the agent's performance. Initially set to 0, the difficulty increases or decreases by 0.1 depending on the success rate over the most recent 20 episodes. We do not update the difficulty again until 10 more episodes have been completed after each adjustment to prevent the difficulty from changing too rapidly. This procedure is presented in Algorithm 1.

We introduced randomness during reset to prevent the robot from forgetting simpler tasks when learning more complex ones. Specifically, the execution difficulty δ_{ex} is sampled from a range $[0, \delta]$, where δ represents the current difficulty. Within this range, a quadratic probability distribution was used, favoring higher-difficulty tasks while still allowing for a small chance of generating easier tasks.

$$P(\delta_{ex}) = \frac{3\delta_{ex}^2}{\delta^3} \tag{1}$$

This ensures the RL agent focuses on learning new tasks while occasionally revisiting previously learned ones.

B. RL for Cable Insertion Tasks

We consider the insertion tasks as a Markov decision process (MDP). The problem is formulated as a tuple $(S,\ A,\ P,\ R,\ \gamma)$, where S is the state space, A is the action space, P defines transition dynamics, R is the reward function, and γ is the discount factor. We chose RLPD [23], an off-policy RL algorithm, to train

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Algorithm 1: Curriculum-Based RL.

Parameters: Curriculum difficulty $\delta = 0$
Success rate of the latest 20 episodes $sr = 0$
Success history of the latest 20 episode $sh = []$
Observation o_t
Output : RL policy π_θ

for $t \leftarrow 0$ *to MAX TIME STEPS* **do**
  Get action a_t from $\pi_\theta(a|o_t)$
  Apply action a_t to the environment and get the reward r_t, the done signal d_t and the next state o_{t+1}
  Add transition (o_t, a_t, o_{t+1}, d_t) to replay buffer
  update $sh \leftarrow d_t$
  if *length(sh)==20* **then**
    $sr \leftarrow$ Compute(sh)
    if $sr < 0.5$ *or* $sr > 0.8$ **then**
      update δ
      Remove 10 elements from the front of sh
    end
  end
  Update π_θ
end

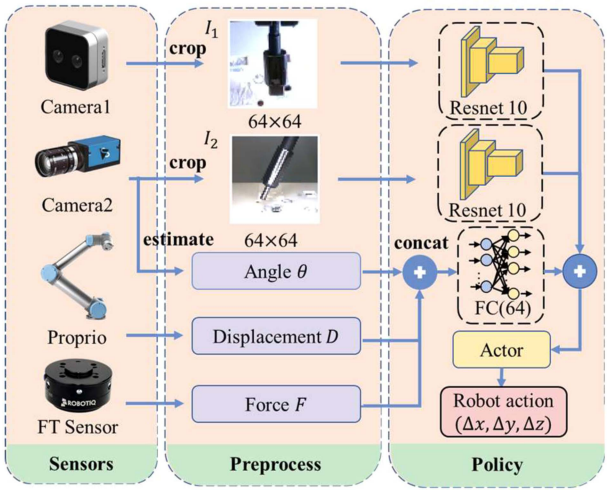


Fig. 3. Overview of the assembly policy. FC is a fully connected layer. The actor, implemented as a fully connected policy network, maps features extracted by the encoder to control actions.

our assembly policy and utilized the implementation provided by [24] to conduct our experiments. RLPD, a variant of the soft actor-critic (SAC) algorithm [25], effectively incorporates both online data and offline demonstration data and performs well in tasks with sparse rewards. RLPD uses symmetric sampling to handle offline data, which means the algorithm samples 50% of the data from the replay buffer and the remaining 50% from the offline data buffer. As part of the data augmentation process, random cropping of images is employed to simulate uncertainty in location. A scheme of the policy is presented in Fig. 3.

1) Design of Action Space: In this task, although there is uncertainty in the cable's bending, as shown in Fig. 1(c), the cable is straightened by translating the robot to create an interaction between the plug and the hole. Since no rotation of the robot is required, the action is defined as $a \in \mathbb{R}^3$, representing the displacement along the x, y, and z axes in the Cartesian coordinate system.

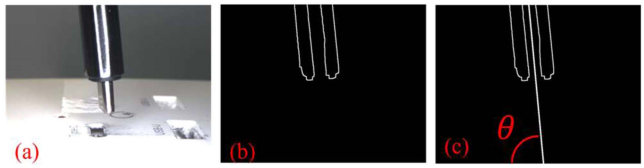


Fig. 4. The angle of the connector estimation. (a) The origin image (b) The contour of the connector (c) The fitted line of the connector's slope.

2) Design of Observation Space: Our approach integrates multiple sensor modalities to capture the robot's state and task-related information. The observation o_t consists of two images I_1, I_2, the displacement D, the three-dimensional force F at the robot end-effector, the angle θ between the connector and the horizontal direction, representing the degree of cable bending, represented as $o_t = (I_1, I_2, D, F, \theta)$.

a) Image: As shown in Fig. 3, two cameras are utilized to capture visual information: an eye-in-hand camera mounted on the robot's wrist and an eye-to-hand camera mounted near the workstation. In this study, RGB images with a resolution of 64 × 64 were used (height(H) = 64, width(W) = 64, and channels(C) = 3). These images are processed through a ResNet10 [26] encoder to extract feature representation.

b) Displacement: To simulate the random initialization of the socket's position without physically relocating it, which would be labor-intensive and impractical in real-world training, the end-effector's displacement D is utilized as the robot's proprioceptive information,

$$D = P_t - P_0 \tag{2}$$

where P_t represents the current position of the end-effector and P_0 represents the initial position after a reset. At the start of each training episode, the robot's end-effector pose is randomly initialized within a specified region of the workspace. From the perspective of the frame attached to the end-effector, this is equivalent to shifting the socket's position.

c) Force: The lateral force applied to the connector causes the cable to bend, limiting the maximum force that can be exerted on it and resulting in minimal torque. Moreover, the torque generated by the eccentricity of the force application point is negligible due to the connector's small size. Therefore, only the three-dimensional directional forces will be provided, i.e., $F = (F_x, F_y, F_z)$.

d) Angle: The angle θ between the connector and the horizontal direction is used to represent the degree of cable bending. As shown in Fig. 4, to calculate θ, images from the eye-to-hand camera are processed using grayscale thresholding and contour extraction to isolate the connector. The least squares method is then applied to fit a straight line $L = \{(x, y)|y = mx + q\}$ to the extracted connector points. θ is the value of the angle as shown below:

$$\theta = \tan^{-1}(m) \tag{3}$$

3) Visual-Based Reward Function: In many assembly studies, the reward function is often designed as a dense function

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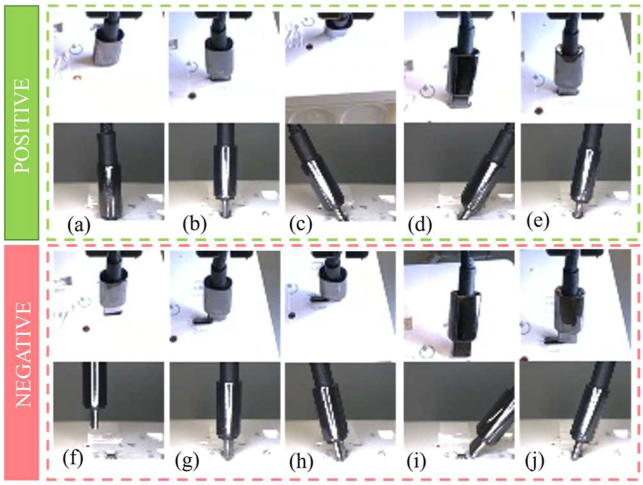


Fig. 5. Training samples for VBRC. Figures (a), (b), (c), (d), (e) show positive samples for VBRC, while Figures (f), (g), (h), (i), (j) depict negative samples. In each figure, the top image is captured by Camera 1, and the bottom image is captured by Camera 2.

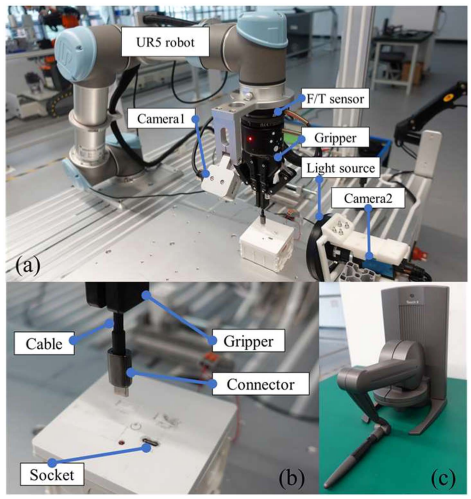


Fig. 6. Overview of the experiment platform.

based on the relative distance between the connector and the socket. However, in our study, effectively utilizing such a reward function is challenging due to localization errors in both the connector and the socket.

To address this challenge, we design a two-phase assembly strategy. Our assembly task is intended to be executed in two phases: a) partially inserting the connector into the socket, and b) adjusting the cable's bending through contact with the socket walls to complete the insertion. In the first phase, a vision-based reward classifier (VBRC) is used to distinguish between two states: the connector not being in the socket and the connector being either partially or fully inserted into the socket. As shown in Fig. 5, it is difficult to determine whether the connector has entered the socket from a single viewpoint. Therefore, the VBRC takes images from two viewpoints as input and uses a convolutional neural network (CNN) to produce a value between 0 and 1, indicating whether the connector is partially inserted. In the second phase, the reward function is defined as a function of the connector's angle, with higher rewards granted for increasing the angle, thereby encouraging the robot to straighten the cable. Finally, task completion is indicated when the socket receives power from the connector. The overall reward function (4) is formulated by combining the rewards from different phases.

$$r = \begin{cases} -0.005 & \text{if } VBRC < 0.5 \\ -0.003 + (\theta - 0.5) * 0.002 & \text{if } VBRC > 0.5 \\ 1 & \text{if done} \end{cases} \quad (4)$$

where θ, the angle of the connector previously mentioned in Section IV-B2, is normalized to a value between 0 and 1 in this context.

C. PD Controller

Sufficient force must be exerted on the connector to overcome the interference fit between the connector and the socket while protecting the cable from excessive external forces that could cause damage. Then, the actions a_t output by the RL agent is processed through a low-level proportional-derivative (PD) controller before being executed by the robot. Unlike traditional methods, the objective is not to keep a constant force contact between the connector and the socket.

Maintaining contact during movement creates frictional forces, leading to cable deformation and making the task more difficult.

The controller adjusts the action a_t output by the policy π_θ to minimize the error E_t, which is the error between the current contact force F_t and the desired contact force F_d.The adjustment Δa is computed using a PD control law (6), and the final action a_{ex} executed by the robot is given in (7).

$$E_t = F_t - F_d \quad (5)$$

$$\Delta a = k_p E_t + k_d \left(E_t - E_{t-1}\right) \quad (6)$$

$$a_{ex} = a_t + \Delta a \quad (7)$$

where k_p and k_d are the proportional and derivative coefficients, respectively. As previously mentioned, F_d is set to the force required to overcome the interference fit, while k_p and k_d are tuned to ensure sensitivity to contact, allowing the robot to move along the socket surface in a bouncing manner.

V. Experiment

We conducted comparisons against existing reinforcement and imitation learning (IL) methods to evaluate the per- formance of the proposed method. Furthermore, ablation studies were designed to analyze the contribution of each component within the framework.

A. Experiment Setup

As shown in Fig. 6, the robot environment setup includes a UR5 [27] robot, a Robotiq FT-300 [28] sensor mounted on the robot end-effector, and a Robotiq 2f-85 gripper [29] to hold the cable. For visual perception, Intel RealSense D405 [30] and Imaging Source DFK 33GX264 [31] are used as eye-in-hand and eye-to-hand cameras, respectively. For data collection, 3D

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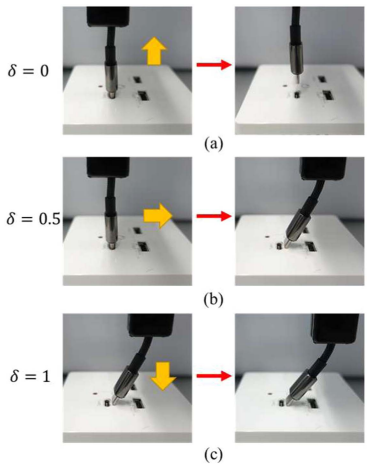


Fig. 7. Reset methods under different curriculum difficulty levels. The yellow arrow indicates the direction of the robot's movement, δ represents difficulty. (a) δ is 0, the plug is directly pulled out. (b) δ is 0.5; the robot moves left or right before pulling out the plug to adjust the cable's bending through collisions. (c) δ is 1; the robot moves downward to achieve greater bending through compression, in addition to the movements at δ 0.5.

Systems TouchX [32] is used as the input interface for human teleoperation of the robot (Fig. 6(c)). Both data collection and model training are conducted on a laptop equipped with an Intel Core i9-11950H CPU and an NVIDIA A5000 GPU.

B. Cable Bending Adjustment Method

A three-stage reset method (Fig. 7) automatically adjusts the initial cable bending. In Stage 1, the connector is reset by being vertically pulled from the socket. In Stage 2, before full removal, the robot moves along the y-axis to cause a collision between the connector and the socket, thereby inducing bending; as difficulty increases, greater displacement results in more pronounced bending. In Stage 3, following Stage 2, the robot moves along the z-axis to apply downward pressure on the cable, further deforming it, with the downward displacement increasing with difficulty. Depending on the task, some stages may be omitted—for example, Stage 3 is skipped at lower difficulty levels. In our setup, the cable is bent nearly 40° from its straight state at the highest difficulty.

This method is efficient and time-saving but relies on connector-socket interaction and is applicable only when the previous episode succeeds. When the cable becomes excessively bent during the hole-searching process due toenvironmental interaction, resetting becomes difficult. Therefore, the cable's state is evaluated before resetting; if the bending exceeds the designed threshold, the cable is reset via interaction with the edge of the working plane. This alternative reset is not standard because it is less efficient than the primary method.

C. VBRC Implementation

Training the VBRC is not laborious, as the demonstration data collected for the RL agent can also be used for VBRC training. During the data collection, the operator assigns binary labels to the data. Instead of labeling each image individually, the operator toggles the label with a trigger, ensuring that all images within a period share the same label until the next switch.

In each transition of the demonstrations, data where the connector is not in the socket is far more prevalent than data where it is. This sample imbalance can be easily addressed by collecting additional positive samples. The classifier is trained before the RL phase, using a total of 3000 images for training.

D. Comparison Experiment

1) Experiment Design: The assembly tasks involve a socket fixed to a worktable and a robot equipped with a gripper holding a Type-C cable approximately 10 mm away from the connector, as shown in Fig. 6(b). To simulate coarse positioning errors of the socket, pose errors within a range of ± 10 mm along the x and y axes are introduced during episode initialization. To comprehensively evaluate the performance of the proposed method, we compare it against several baseline approaches, including RL and IL methods. All comparison groups are listed as follows:

a) **Our Method:** As described earlier, the proposed method combines RL with demonstration data. We collected trajectory data for the assembly process through teleoperation, acquiring 20 episodes for each of three difficulty levels (0, 0.5, and 1), resulting in a total of 60 demonstration trajectories.

b) **LBSTM** [12]: This method employs RL combined with CL, where the action space is expanded progressively from simple translations to larger rotations. Additionally, it enhances training efficiency by utilizing pixel data near the connector instead of processing the full image. Unlike the original approach, where the gripped object is fixed relative to the gripper, our implementation introduces cable bending into the CL.

c) **ACT** [33]: ACT is one of the most recent and effective methods in IL for robotics. In our implementation, the observation space was aligned with that of our method, and we similarly collected 20 episodes for each of the three difficulty levels. A higher sampling frequency was adopted to meet the training requirements of IL algorithms, resulting in a dataset approximately eight times larger than that used in our method.

2) Result and Analysis: In this section, we evaluate the aforementioned methods based on test results from three difficulty levels, including success rate (SR) and average time (AT). Each difficulty level was tested 50 times. Tasks that reach the maximum timestep without completion are considered failures. The results are presented in Table I.

After 3 hours of training, our method outperformed others across all difficulty levels, achieving a 98% success rate even at the highest difficulty. It also demonstrated shorter completion times compared to the other methods. As task difficulty increased, completion time also increased, supporting the assumption that cable bending significantly affects task difficulty. At the highest difficulty, completion time increases significantly due to excessive contact forces during cable bending adjustments, occasionally causing the PD controller to push the plug upward and out of the socket.

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TABLE I
EXPERIMENTAL RESULTS OF DIFFERENT METHODS

Method	Difficulty	SR [%]	AT [s]
Ours	0	**100**	1.7
	0.5	**100**	**1.8**
	1	**98**	**3.23**
LBTSM [12]	0	96	1.65
	0.5	0	-
	1	0	-
ACT [33]	0	72	3.45
	0.5	54	3.63
	1	52	5.13

LBSTM, trained for 3.5 hours, showed success only at difficulty level 0. It is efficient for tasks with minimal changes in connector-gripper position, but struggles with more complex tasks, indicating that simple pixel observations are insufficient.

The ACT method was trained for 5 hours, and some progress was made across all difficulty levels. Its success highly depends on the initial state, including the degree of cable bending and the robot's initial position. This indicates that the demonstrations collected may not fully cover all task states. The longer completion time of this method is mainly due to slower decision-making, caused by two factors: ACT computes actions over multiple timesteps, while RL methods calculate only the next action, and the higher sampling frequency during data collection results in more minor actions, requiring more steps to cover the same distance.

In summary, LBTSM is restricted to the assembly of rigid objects. Additionally, the ACT approach demands extensive demonstrations. In contrast, our method produces strong results with fewer demonstrations.

E. Ablation Experiment

1) Experiment Design: The assembly task in this section is identical to that described in the previous section. All experiments were repeated 5 times using different random seeds. The ablation experiments are as follows:

a) **No Curriculum Learning (NCL):** This method does not use CL; instead, it randomly adjusts the degree of cable bending at the beginning of each episode, utilizing interaction with the edge of the working plane for adjustment, as mentioned in V.B.

b) **No Demonstration (ND):** This method does not use expert demonstrations as training data for RL; however, it still utilizes the reward function trained on those demonstrations.

c) **Sparse Reward (SR):** This method relies on a sparse reward, which is defined as follows:

$$r = \begin{cases} -0.005 & \text{if not done} \\ 1 & \text{if done} \end{cases} \tag{8}$$

d) **No Angle (NA):** this method excludes the angle from the observation space but retains it in the reward function.

e) **No Displacement (NDI):** this method excludes the displacement from the observation space.

2) Result and Analysis: This section analyzes the training process, focusing on the success rate and its variation with task difficulty under different ablation settings. Each experiment was trained for 100000 steps, taking approximately 3 hours. The results are shown in Fig. 8.

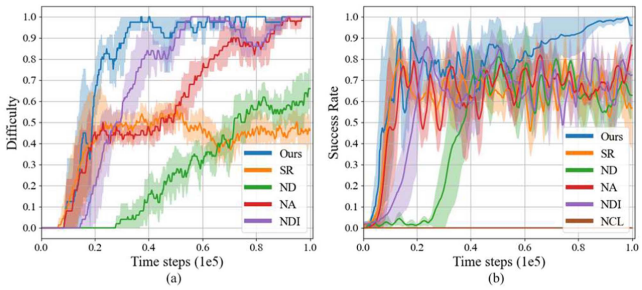


Fig. 8. Training process of the ablation experiments. (a) Difficulty variation curve, (b) Success rate variation curve.

Our method is the only one that has achieved the highest difficulty level and a high success rate. Although the highest difficulty is reached after 30000 steps, a high success rate is observed only around 80000 steps. Despite implementing a mechanism to slow down difficulty updates, the results indicate that the agent remains undertrained at lower difficulty levels before the difficulty increases. Overall, the training efficiency of the other ablation methods is lower than ours.

The SR method performs similarly to ours during early training but struggles to increase difficulty beyond 0.5. This indicates that without a well-designed reward function, the RL agent struggles to master cable bending adjustment.

The ND method maintains a constant difficulty level for the first 30000 steps, only increasing it thereafter, yet fails to achieve the highest difficulty level by the end of training. Moreover, it requires more time to master low-difficulty tasks, and the progression of task difficulty is significantly slower. These results underscore the critical role of demonstrations in enhancing training efficiency.

The NA method performs similarly to the SR method until 40000 steps. However, the lack of angle information delays progress. Improvement emerges after 50000 steps, indicating that image inputs alone can suffice. Still, low-dimensional angle data offers a more direct state representation, likely expediting training.

The NDI method shows only a minor performance gap relative to our approach. It requires approximately 8000 additional steps to master the easiest tasks and reach the highest difficulty level at around 50000 steps, indicating a slightly slower training pace. Nevertheless, although displacement does not directly encode the plug-socket relative position, it alleviates the challenge of inferring precise positions solely from image data.

The difficulty of the NCL method is not plotted since it is generated randomly. The success rate curve reveals that the method is unsuccessful, suggesting that the RL agent struggles to learn the task in a highly complex space.

VI. CONCLUSION

In this study, we have proposed a curriculum reinforcement learning-based framework to accomplish the insertion task of DLOs by utilizing the deformations that occur upon interaction

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with the environment. We combined a vision-based reward function with CL, enabling the RL agent to progress from simple to complex tasks. This approach allowed us to achieve training in a high-dimensional task space using a relatively sparse reward function.

Our results show that the generated policy can successfully handle assembly tasks involving cables bent up to 40° from their straight state, achieving high success rates and rapid completion times compared to existing approaches. The ablation experiments confirm that the designed reward function, CL, collected expert demonstrations, and observation designs played crucial roles.

This method can complete the insertion task with varying degrees of cable bending, but it requires the grasped cable's orientation to be known. Furthermore, the fixed camera setup limits the method's adaptability. Future work will focus on integrating tactile sensors for insertion tasks with varying grasping orientations, and more flexible image processing methods should be considered.

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